

VALUING CORPORATE SECURITIES: SOME EFFECTS OF BOND INDENTURE PROVISIONS

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I. THE VALUATION OF CORPORATE SECURITIES

IN A RECENT PAPER Black and Scholes [3] presented an explicit equilibrium model for valuing options. In this paper they indicated that a similar analysis could potentially be applied to all corporate securities. In other papers, both Merton [8] and Ross [11] noted the broad applicability of option pricing arguments. At the same time Black and Scholes also pointed out that actual security indentures have a variety of conditions that would bring new features and complications into the valuation process.

Our objective in this paper is to make some general statements on this valuation process and then turn to an analysis of certain types of bond indenture provisions which are often found in practice. Specifically, we will look at the effects of safety covenants, subordination arrangements, and restrictions on the financing of interest and dividend payments.

Throughout the paper we will make the following assumptions:

- a1) Every individual acts as if he can buy or sell as much of any security as he wishes without affecting the market price.
- a2) There exists a riskless asset paying a known constant interest rate r .
- a3) Individuals may take short positions in any security, including the riskless asset, and receive the proceeds of the sale. Restitution is required for payouts made to securities held short.
- a4) Trading takes place continuously.
- a5) There are no taxes, indivisibilities, bankruptcy costs, transaction costs, or agency costs.
- a6) The value of the firm follows a diffusion process with instantaneous variance proportional to the square of the value.

This last assumption is quite important and needs some amplification. Until very recently this was the standard framework for discussions of contingent claim pricing. Increasing evidence, however, indicates that it may not be completely appropriate.¹ The instantaneous variance may be some other function of the firm

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1. See Black [1].

value, and possibly dependent on time as well.² It may also depend on other random variables. Furthermore, discontinuities associated with jump processes may be important.³ Nevertheless, this assumption provides a useful setting for the points we want to make and facilitates comparison with earlier results.

With these assumptions, the standard hedging or capital asset pricing arguments lead to a valuation equation. For the process we are considering here, it is derived in its most general form in Merton [9] as

$$\frac{1}{2}\sigma^2 V^2 f_{vv} + (rV - p(V, t))f_v - rf + f_t + p'(V, t) = 0 \quad (1)$$

where f is a generic label for any of the firm's securities, V is the value of the firm, t denotes time, σ^2 is the instantaneous variance of the return on the firm, $p(V, t)$ is the net total payout made, or inflow received, by the firm, and $p'(V, t)$ is the payout received or payment made by security f .

Suppose the firm has outstanding only equity and a single bond issue with a promised final payment of P . At the maturity date of the bonds, T , the stockholders will pay off the bondholders if they can. If they cannot, the ownership of the firm passes to the bondholders. So at time T , the bonds will have the value $\min(V, P)$ and the stock will have the value $\max(V - P, 0)$.

Now this formulation already implicitly contains several assumptions about the bond indenture. The fact that σ^2 , $p(V, t)$ and $p'(V, t)$, and P were assumed known (and finite) implies that the bond contract renders them determinate by placing limiting restrictions on, respectively, the firm's investment, payout, and further financing policies.

Furthermore, it assumes that the fortunes of the firm may cause its value to rise to an arbitrarily high level or dwindle to nearly nothing without any sort of reorganization occurring in the firm's financial arrangements. More generally, there may be both lower and upper boundaries at which the firm's securities must take on specific values. The boundaries may be given exogenously by the contract specifications or determined endogenously as part of an optimal decision problem.

The indenture agreements which we will consider serve as examples of a specified or induced lower boundary at which the firm will be reorganized. An example of an upper boundary is a call provision on a bond.⁴ Also, the final payment at the maturity date may be a quite arbitrary function of the value of the firm at that time, $\xi(V(T))$.

It will be helpful to look at this problem in a way discussed in Cox and Ross [5, 6].⁵ The valuation equation (1) does not involve preferences, so a solution derived for any specific set of preferences must hold in general. In particular, the

2. See Cox and Ross [6] for a discussion of some models of this type.

3. Processes with discontinuous sample paths are examined in Cox and Ross [5, 6] and Merton [10].

4. Call provisions on bonds have recently been examined by Brennan and Schwartz [4] and Ingersoll [7]. All of our results could be extended to include such upper boundaries as well.

5. For a related discussion, see Black and Scholes [2].

relative value of contingent claims in terms of the value of underlying assets must be consistent with risk neutrality.⁶

If we know the distribution of the underlying assets in a risk-neutral world, then we can readily solve a number of valuation problems.⁷ We can in our problem think of each security as having four sources of value: its value at the maturity date if the firm is not reorganized before then, its value if the firm is reorganized at the lower boundary, its value if the firm is reorganized at the upper boundary, and the value of the payouts it will potentially receive. Although the first three sources are mutually exclusive, they are all possible outcomes given our current position, so they each contribute to current value. The contribution to the total value of a claim of any of its component sources will in a risk neutral world simply be the discounted expected value of that component.

For any claim f , let $h_i(V(t), t)$, $i = 1, \dots, 4$, denote respectively the four components referred to above, so $f(V(t), t) = \sum_{i=1}^4 h_i(V(t), t)$. Let $g_1(\tau)$ ($g_2(\tau)$) be the value of f , as given by the contract, if the firm is reorganized at the lower (upper) boundary $C_1(\tau)$ ($C_2(\tau)$) at time τ . Denote the distribution in a risk neutral world of the value of the firm at time τ , $V(\tau)$, conditional on its value at the current time t , $V(t)$, $C_1(t) < V(t) < C_2(t)$, as $\Phi(V(\tau), \tau | V(t), t)$. Then taking the indicated expectations we can write

$$h_1(V(t), t) = e^{-r(T-t)} \int_{\kappa(T)} \xi(V(T)) d\Phi(V(T), T | V(t), t) \quad (2)$$

and

$$h_4(V(t), t) = \int_t^T e^{-r(s-t)} \left[\int_{\kappa(s)} p'(V(s), s) d\Phi(V(s), s | V(t), t) \right] ds, \quad (3)$$

where $\kappa(\cdot)$ denotes the interval $(C_1(\cdot), C_2(\cdot))$.

The contribution of the potential value at the reorganization boundaries is somewhat different. Formerly we knew the time of receipt of each potential payment but not the amount which would actually be received. Here the amount to be received at each boundary is a known function specified by the contract, but the time of receipt is a random variable. However, its distribution is just that of the first passage time to the boundary, and the approach taken by Cox and Ross can still be applied.

6. The ability to form a perfectly hedged portfolio is a sufficient condition for the derivation of a valuation equation free of preferences. Note that this does not say that the value of the underlying assets in terms of the values of other assets is independent of preferences.

7. In a risk neutral world the instantaneous mean total return must be rV , so the instantaneous mean of the price component must be $rV - p(V, t)$. For a diffusion process, this, together with the instantaneous variance and behavior at accessible boundaries, completely specifies the processes. The value of the assets of the firm would in general have only a lower barrier, an absorbing one at the origin. However, our interest is in probabilities for paths of firm value which have not previously reached one of the reorganization boundaries. A convenient way to introduce this is by considering the distribution with the boundaries taken as artificial absorbing barriers, and we will adopt this convention.

Let $\Psi_1(t^*|V(t),t)$ be the distribution of the first passage time t^* to the lower boundary, and let $\Psi_2(t^*|V(t),t)$ denote the corresponding distribution for the upper boundary. Then

$$h_{i+1}(V(t),t) = \int_t^T e^{-r(t^*-t)} g_i(t^*) d\Psi_i(t^*|V(t),t), \quad i=1,2. \quad (4)$$

This development also disposes of uniqueness problems, since economically inadmissible solutions to the valuation equation are automatically avoided by the probabilistic approach. However, it cannot be applied directly to situations where the boundaries must be determined endogenously as part of an optimal stopping problem.

Actual payouts by firms, of course, occur in lumps at discrete intervals. In many situations it is more convenient and perfectly acceptable to represent these payouts as a continual flow. Many other times, however, it is preferable to explicitly recognize the discrete nature of things. This is particularly true in optimal stopping problems when the structure of the problem dictates that decisions will be made only at these discrete points. An example in terms of options would be an American call on a stock paying discrete dividends. Restrictions on the financing of coupon payments to debt, which we will discuss later, provides an example in terms of corporate liabilities. To solve these problems we could work recursively, with the terminal condition at each stage determined by the solution to the previous stage. Start at the last payment date. If a decision is made to stop at this point, the claimholder receives a payoff given by the terms of the contract. If he does not stop, his payoff is the value of a claim with one more period to go, given that the value of the firm is its current value minus the payment. This value is determined by the payment to be received at the maturity date. The claimholder can then determine his optimal decision rule. With the optimal decision rule specified, we can find the value of the claim as a function of firm value at the last decision point. At the next-to-last decision point we would face an identical problem except that the value function we just found would take the place of the function giving the payment to be received at the maturity date. By working backward we can find the value of the claim at any time. Note that this gives only an approximate solution when the optimal decision points are actually continuous in time. However, we could always get a better approximation by adding more discrete decision points, even though no payouts are being made at these additional points.

Throughout the paper we will make use of the relationship between the equilibrium expected return on any of the individual securities of the firm, ν , and the (exogenously determined) equilibrium expected return on the total firm, μ . As given in Black and Scholes [3] and Merton [9], this is $\nu - r = (Vf_v/f)(\mu - r)$. Furthermore, since the process followed by any individual security is a transformation of that governing the total value of the firm, its instantaneous variance will be $\sigma^2 V^2 [f_v]^2$. Thus we can write the ratio of the instantaneous standard deviation of the rate of return on any individual security to that of the firm as Vf_v/f . Another way to say this is that in equilibrium the excess expected return per unit of risk must be the same for all of the firm's securities. The elasticity Vf_v/f thus conveys the essential

information about relative risk and expected return. In subsequent use of the term elasticity, we will always be referring to this function.

II. BONDS WITH SAFETY COVENANTS

In this section we will consider the effects of safety covenants on the value and behavior of the firm's securities. Safety covenants are contractual provisions which give the bondholders the right to bankrupt or force a reorganization of the firm if it is doing poorly according to some standard. One standard for this may be the omission of interest payments on the debt. However, if the stockholders are allowed to sell the assets of the firm to meet the interest payments, then this restriction is not very effective. In this situation a natural form for a safety covenant is the following: if the value of the firm falls to a specified level, which may change over time, then the bondholders are entitled to force the firm into bankruptcy and obtain the ownership of the assets. In this form of agreement, interest payments to the debt do not play a critical role, so we will assume that the firm has outstanding only a single issue of discount bonds. We will, however, assume that the contractual provisions allow the stockholders to receive a continuous dividend payment, aV , proportional to the value of the firm. With a continuous time analysis, it is quite reasonable for the time dependence of the safety covenant to take an exponential form, so we will let the specified bankruptcy level, $C_1(t)$, be $Ce^{-\gamma(T-t)}$.

The relevant form of the valuation equation (1) for the bonds, B , will be

$$\frac{1}{2}\sigma^2V^2B_{vv} + (r-a)VB_v - rB + B_t = 0 \quad (5)$$

with boundary conditions

$$B(V, T) = \min(V, P)$$

$$B(Ce^{-\gamma(T-t)}, t) = Ce^{-\gamma(T-t)}.$$

Similarly, the value of the stock, S , must satisfy

$$\frac{1}{2}\sigma^2V^2S_{vv} + (r-a)VS_v - rS + S_t + aV = 0 \quad (6)$$

with boundary conditions

$$S(V, T) = \max(V - P, 0)$$

$$S(Ce^{-\gamma(T-t)}, t) = 0.$$

To apply the probabilistic approach to valuation we need $\Phi(V(\tau), \tau | V(t), t)$, the distribution in a risk neutral world of the value of the firm at time τ , $V(\tau)$, conditional on its value at the current time t , $V(t) = V$. Under our assumptions, this will be the distribution of a lognormal process with an (artificial) absorbing barrier at the reorganization boundary $C_1(\tau) = Ce^{-\gamma(T-\tau)}$. The probability that $V(\tau) \geq K$

and has not reached the reorganization boundary in the meantime is given by

$$N\left(\frac{\ln V - \ln K + (r - a - \frac{1}{2}\sigma^2)(\tau - t)}{\sqrt{\sigma^2(\tau - t)}}\right) - \left(\frac{V}{Ce^{-\gamma(T-t)}}\right)^{1-(2(r-a-\gamma)/\sigma^2)} N\left(\frac{2\ln Ce^{-\gamma(T-t)} - \ln V - \ln K + (r - a - \frac{1}{2}\sigma^2)(\tau - t)}{\sqrt{\sigma^2(\tau - t)}}\right), \quad (7)$$

where $N(\cdot)$ is the unit normal distribution function. Setting $K = Ce^{-\gamma(T-\tau)}$ gives the probability in a risk neutral world that the firm has not been reorganized before time τ . This is the complementary first passage time distribution. That is, if t^* is the first passage time to the boundary, the probability that $t^* \geq \tau$ is obtained from (7) by letting $K = Ce^{-\gamma(T-\tau)}$.

By using these distributions to find the expected discounted value of the payments, we can obtain the valuation formula for B as

$$B(V, t) = Pe^{-r(T-t)}[N(z_1) - y^{2\theta-2}N(z_2)] + Ve^{-a(T-t)}[N(z_3) + y^{2\theta}N(z_4) + y^{\theta+\zeta}e^{a(T-t)}N(z_5) + y^{\theta-\zeta}e^{a(T-t)}N(z_6) - y^{\theta-\eta}N(z_7) - y^{\theta-\eta}N(z_8)], \quad (8)$$

where

$$y = Ce^{-\gamma(T-t)} / V$$

$$\theta = (r - a - \gamma + \frac{1}{2}\sigma^2) / \sigma^2$$

$$\delta = (r - a - \gamma - \frac{1}{2}\sigma^2)^2 + 2\sigma^2(r - \gamma)$$

$$\zeta = \sqrt{\delta} / \sigma^2$$

$$\eta = \sqrt{\delta - 2\sigma^2 a} / \sigma^2$$

$$z_1 = [\ln V - \ln P + (r - a - \frac{1}{2}\sigma^2)(T - t)] / \sqrt{\sigma^2(T - t)}$$

$$z_2 = [\ln V - \ln P + 2\ln y + (r - a - \frac{1}{2}\sigma^2)(T - t)] / \sqrt{\sigma^2(T - t)}$$

$$z_3 = [\ln P - \ln V - (r - a + \frac{1}{2}\sigma^2)(T - t)] / \sqrt{\sigma^2(T - t)}$$

$$z_4 = [\ln V - \ln P + 2\ln y + (r - a + \frac{1}{2}\sigma^2)(T - t)] / \sqrt{\sigma^2(T - t)}$$

$$z_5 = [\ln y + \zeta\sigma^2(T - t)] / \sqrt{\sigma^2(T - t)}$$

$$z_6 = [\ln y - \zeta\sigma^2(T - t)] / \sqrt{\sigma^2(T - t)}$$

$$z_7 = [\ln y + \eta\sigma^2(T - t)] / \sqrt{\sigma^2(T - t)}$$

$$z_8 = [\ln y - \eta\sigma^2(T - t)] / \sqrt{\sigma^2(T - t)}.$$

This formula holds for all $Ce^{-\gamma(T-t)} \leq Pe^{r(T-t)}$. An interesting choice is $Ce^{-\gamma(T-t)} = \rho Pe^{-r(T-t)}$, with $0 \leq \rho \leq 1$, so that the reorganization value specified in the safety covenant is a constant fraction of the present value of the promised final payment. For clarity in making comparisons, we will use only this form below.

Merton [9] has extensively studied in this setting the properties of discount bonds when there are no safety covenants and no dividends. Rather than repeat parts of his analysis, we will focus on properties which are particular to the existence of safety covenants. The most basic properties, such as the fact that B is an increasing function of V and t and a decreasing function of σ^2 , r , and a remain the same.

It is easy to verify that B is an increasing function of ρ . Contrary to what is sometimes claimed, premature bankruptcy is not in itself detrimental for the bondholders. It is in their interests to have a contract which will force bankruptcy as quickly as possible. If bankruptcy occurs, the total ownership of the firm will pass to the bondholders, and this is the best they can achieve in any circumstances. A second look shows that B is a convex function of ρ , going to $Pe^{-r(T-t)}$, the riskless value, as ρ goes to one. The elasticity of B is a decreasing concave function of ρ , going to zero as ρ goes to one, so a higher bankruptcy level always makes the debt safer. The elasticity of the stock is an increasing convex function of ρ .

Safety covenants provide a floor value for the bond which limits the gains to stockholders from somehow circumventing the other indenture restrictions. For example, as either σ^2 or a goes to infinity, the value of the bonds goes to $\rho Pe^{-r(T-t)}$ rather than zero. Similarly, if we compare the riskiness of bonds of firms differing only in investment policy or dividend policy, we find important differences for large values of a and σ^2 . If $\rho=0$, the elasticity is an increasing concave function of a , going to one as a goes to infinity. If $\rho>0$, the elasticity has an initial increasing concave segment, but then reaches a maximum, followed successively by decreasing concave and convex segments going to zero as a goes to infinity. The behavior of the elasticity with respect to the variance is for small values of σ^2 qualitatively the same as the case with no safety covenant, but as σ^2 becomes large, it approaches zero rather than one-half.

The behavior of the elasticities with respect to the value of the firm is also interesting and is shown in Figure 1. When the stock is entitled to receive dividends, as the value of the firm declines, we find that the riskiness and expected return of the stock first increases, then decreases, and finally increases again as the value approaches the bankruptcy boundary. Intuitively we could think of this in the following way. For values of V near the boundary it is quite likely that the stockholders will lose everything and their claim is accordingly quite risky. As V increases, we reach a stage where bankruptcy is no longer imminent, but it is most unlikely that anything will be left for the stockholders at the maturity date. The value of the stock derives almost solely from the value of the dividends it is entitled to receive, and these are proportional to the value of the firm and hence have unitary elasticity. As V increases further, the major part of the stock's value becomes due to the uncertain amount it may receive at the maturity date, and hence the riskiness increases. Finally, as V reaches a very high level, it becomes virtually certain that the bonds will be redeemed in full and the stock becomes

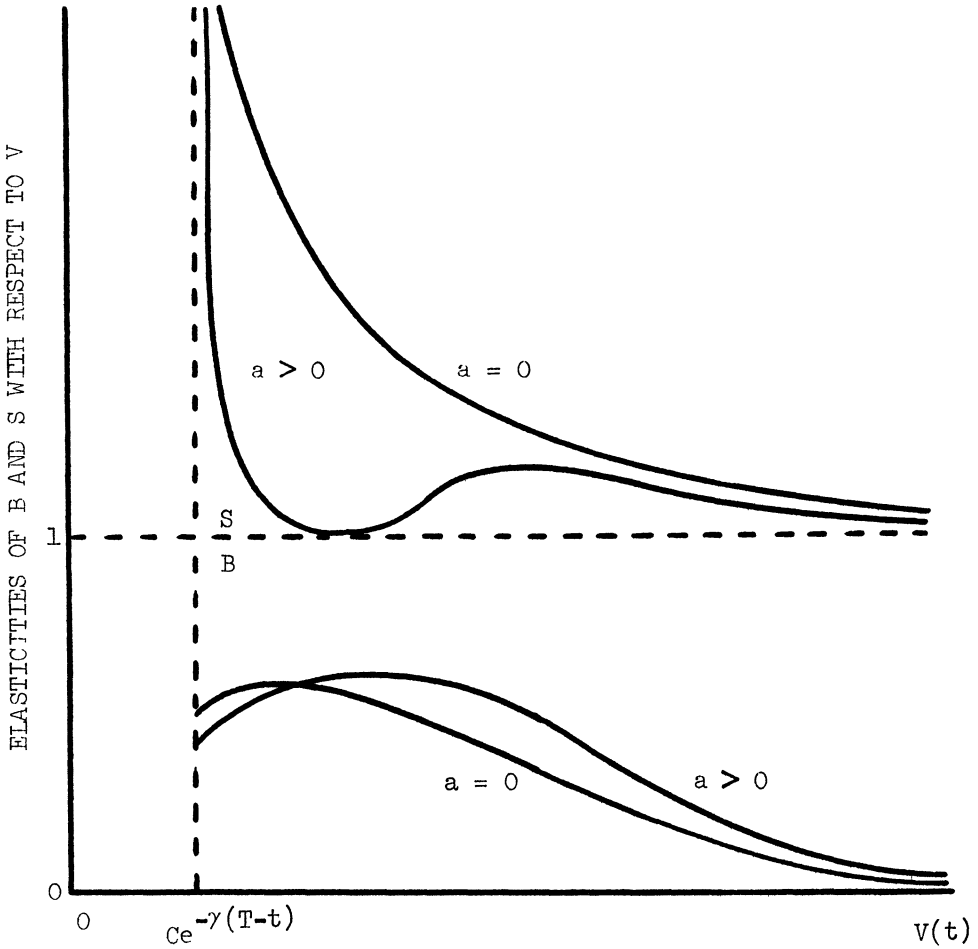


FIGURE 1 Current Value of the Firm

equivalent to a levered position in the firm as a whole, with degree of leverage $V/(V - Pe^{-r(T-t)})$.

III. SUBORDINATED BONDS

Another common form of indenture agreement involves the subordination of the claims of one class of debt holders, the junior bonds, to those of a second class, the senior bonds. At the maturity date of the bonds, payments can be made to the junior debt holders only if the full promised payment to the senior debt holders has been made. Suppose that both classes of bonds are discount bonds, and let the promised payments to senior and junior debt be, respectively, P and Q . Then at the maturity date the value of each of the firm's securities will be as shown in Table 1.

This problem could be solved separately by the methods used earlier, but this is unnecessary since we can write the solution in terms of (8). To see this, note that

TABLE 1

VALUES OF CLAIMS AT MATURITY			
Claim	$V < P$	$P \leq V \leq P + Q$	$V > P + Q$
Senior Bonds	V	P	P
Junior Bonds	0	$V - P$	Q
Stock	0	0	$V - P - Q$

the value of the senior bond (or stock) is the same as the corresponding security of an identical firm with a single bond issue having a promised payment of P (or $(P + Q)$). Let $B(V, t; P, \rho Pe^{-r(T-t)})$ denote the formula given in (8) for a single bond issue with promised payment P and a safety covenant boundary given by $\rho Pe^{-r(T-t)}$. Then the value of the junior debt, J , can be written as

$$\begin{aligned}
 J(V, t) &= B(V, t; P + Q, \rho Pe^{-r(T-t)}) - B(V, t; P, \rho Pe^{-r(T-t)}), \quad \rho < 1 \\
 &= B(V, t; P + Q, \rho Pe^{-r(T-t)}) - Pe^{-r(T-t)}, \quad 1 \leq \rho \leq \frac{P+Q}{P} \\
 &= Qe^{-r(T-t)}, \quad \rho > \frac{P+Q}{P}.
 \end{aligned} \tag{9}$$

The discussion in the first section suggested that the values of junior and senior discount bonds, and correspondingly of options with different exercise prices, could be given a geometric interpretation. Consider the case with no payouts and no safety covenants. Depict graphically the distribution function $\Phi(V(T), T | V(t), t)$. Then as shown in Figure 2, the values of the firm's securities can be interpreted as areas above the distribution function, when these areas are multiplied by the discount factor $e^{-r(T-t)}$.

To see this, consider, for example, the senior bonds. Since

$$\int_0^P V(T) d\Phi = \int_0^P [1 - \Phi(V(T))] dV(T) - P[1 - \Phi(P)],$$

then

$$\int_0^\infty \min(V(T), P) d\Phi = \int_0^P [1 - \Phi(V(T))] dV(T),$$

which is represented by the indicated area.

Subordination does indeed achieve its anticipated effect of giving the senior bonds a larger value than they would have if they were the corresponding fraction

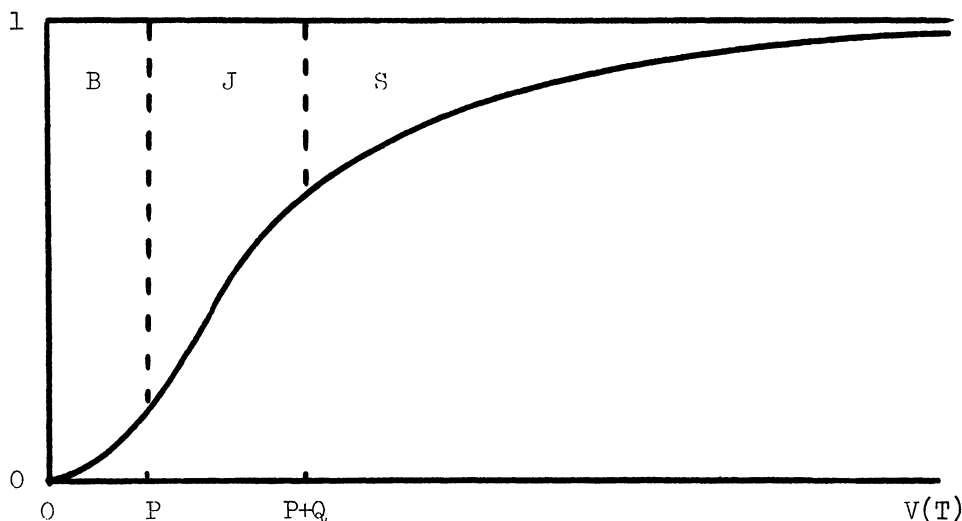


FIGURE 2 Security Values as Areas Above the Distribution Function

of an undifferentiated bond issue. That is, the value of the senior bonds will be greater than $P/(P+Q)$ times the value of a single issue with promised payment $P+Q$. This follows directly from the concavity of discount bonds in the final payment.

The effects of a safety covenant on the subordinated debt are just as we would expect. J is initially a decreasing convex function of ρ , reaching a minimum when $\rho=1$. For $\rho>1$, it is an increasing convex function, reaching a maximum when $\rho=P+Q$. For values of $\rho<1$, the benefits of the safety covenant accrue entirely to the senior bondholders and are partly at the expense of the junior bondholders as well as the stockholders. As ρ increases, the junior bondholders begin to receive benefits as well, and finally the entire expense falls upon the stockholders. In the remainder of this section we will let $\rho=0$.

Further analysis shows that the subordinated debt has many characteristics which are quite different from those normally associated with bonds. While senior bonds are always a concave function of V , the junior bonds are initially a convex function of V , becoming a concave function for larger values of V . The inflection point, V^* , occurs at

$$V = [P(P+Q)]^{1/2} \exp[-(r-a+\frac{1}{2}\sigma^2)(T-t)]. \quad (10)$$

Again unlike the senior debt, the value of the junior debt can be an increasing function of σ^2 . Analysis of the function shows J is an increasing (decreasing) function of σ^2 for V less than (greater than) V^* . This means that the bondholders as a group may under some circumstances have conflicting interests with respect to changes in the total riskiness of the firm's investment policy. To fully protect the value of their claims, the senior bondholders must insist on the sole right to approve investment policy changes which will increase the business risk of the firm.

As we might now expect, J can be an increasing function of time to maturity. Unlike the senior debt, it is possible for the junior debt to be worthless at maturity, and if such a development is imminent, the junior bondholders would find it in their interests to try to extend the maturity date of the entire bond issue. Although it is possible for the value of the junior bonds to be either a decreasing or increasing function of the interest rate, it is always a decreasing function of the dividend rate.

Turning now to the characteristics of risk and expected return, we find that the junior bonds behave partly like a senior bond and partly like a stock. We normally think of a bond as being less risky than the assets of the firm, that is, having an elasticity of less than one, and of the stock as being more risky than the assets. However, we find that the elasticity, ϵ , of J is a decreasing convex function of V which goes to zero as V goes to infinity and to infinity as V goes to zero. Further inspection shows that

$$\epsilon \begin{cases} > \\ < \end{cases} 1 \quad \text{as} \quad PN(z_1; P) \begin{cases} > \\ < \end{cases} (P+Q)N(z_1; P+Q). \quad (11)$$

The behavior of the elasticity with respect to time until maturity for the relevant firm and parameter values is shown in Figure 3.

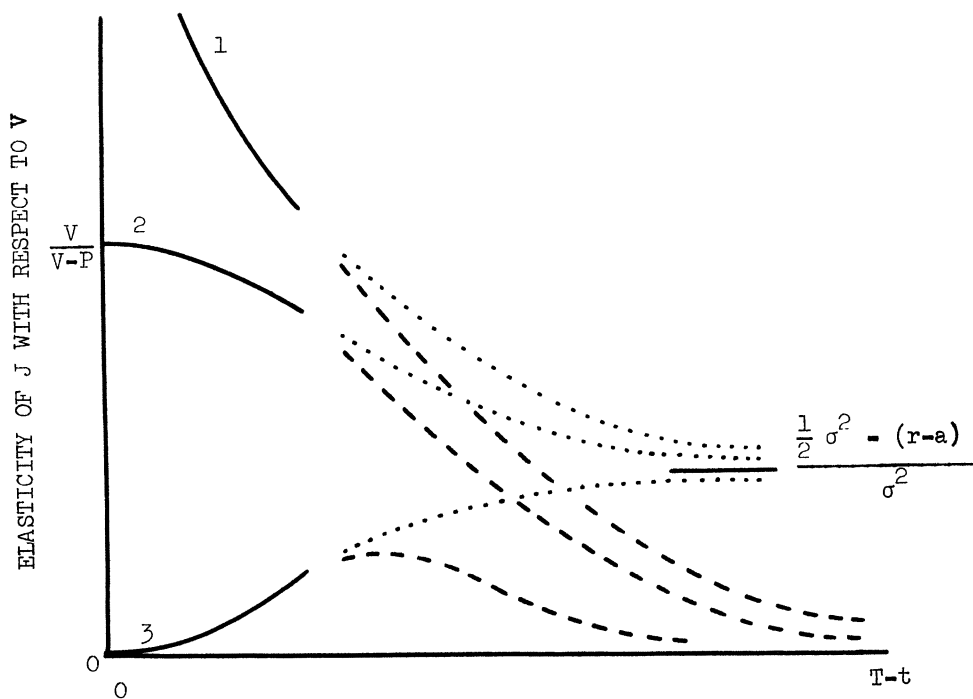


FIGURE 3 Time Until Maturity
 $1 = V < P$, $2 = P < V < P + Q$, $3 = P + Q < V$
 $\dots\dots = (r-a) < \frac{1}{2}\sigma^2$, $\text{----} = (r-a) \geq \frac{1}{2}\sigma^2$

IV. RESTRICTIONS ON THE FINANCING OF INTEREST AND DIVIDEND PAYMENTS

Suppose now that the firm has interest paying bonds outstanding. In this section we will see that it is quite important how the stockholders are allowed to raise the money to make the payments to the bondholders. Previous studies of interest paying bonds have assumed that the stockholders are allowed to sell the assets of the firm to make these payments. Many bonds have contractual provisions which limit the extent to which this can be done. To focus on the effects of these restrictions, suppose that the sale of assets for this purpose is in fact completely forbidden. Interest payments, and any dividend payments, must be financed by issuing new securities. To protect the value of their claim the bondholders must also require that the new securities be equity or subordinated bonds.

For concreteness suppose the bonds have a promised final payment of P and make periodic interest payments of $c = Pe^{rt'}$, where t' is the interval between payments. If an interest payment is not made, the firm is in default and the promised payment P becomes due immediately. The bonds would then be worth $\min(V, P + c)$. Since this is the maximum value the bonds can possibly have, the bondholders would always be glad to see a payment missed, and correspondingly the stockholders would always want to make the payment if there is any way they possibly can. However, they may not be able to. This would happen whenever the value of the equity after the payment is made, if it is made, would be less than the value of the payment. Even if the present stockholders offered an equity issue which would dilute their own interest to virtually nothing, they would still find no takers for it. All of this can occur when the assets of the firm still have substantial value. It provides one explanation, along with the safety covenants discussed earlier, of the observed fact that many firms end up in bankruptcy and reorganization even though their total value may be quite significant.

Under these conditions the use of junior debt, and the exact terms of the junior debt, have important implications. Suppose that because of legal restrictions or diffusion of ownership the junior bondholders are forced to play a purely passive role. They cannot at some later date agree to a change in their contract or take an active part in the firm. To protect themselves in these circumstances, the junior bondholders must require that any subsequent debt issues be subordinated to their own.

However, issuing any junior debt at all in this situation would actually help the senior bondholders and hurt the stockholders. This is because it would then be more likely that a payment will be missed and the bondholders will take over the firm. To see this, consider the value of the claims after a payment has been made. In an attempt to raise the money to in fact make that payment the stockholders were formerly able to offer up for sale the total value of the firm less the value of the senior bonds, while now they can offer only the total value less the value of both the senior and junior bonds. The senior bondholders would be better off, and assuming that the junior debt was sold at a fair price, the difference would have to come out of the pockets of the stockholders.

If it is possible for the junior debtholders to subsequently voluntarily change their status, things will be different. They may find it in their interests to permit the

issue of additional unsubordinated debt rather than allow a payment to be missed. In fact, the disadvantages of junior debt could be completely circumvented by a contract of the following kind. Suppose that in the junior debt indenture it is specified that if the stockholders find that they cannot make a payment by issuing new equity, they will sign their entire equity interest over to the junior bondholders. The junior bondholders could then immediately reorganize the firm as one having only equity and senior bonds. If such an arrangement is possible, there would then be no disadvantage to issuing junior debt, since the firm would in effect switch back to equity at exactly the moment the debt would have been a disadvantage.

We have stated the discussion in terms of extreme cases in order to highlight the issues. Often there may be only partial restrictions on the sale of assets, such as those allowing the sale of assets added by current earnings, or the junior bondholders may be able to partly change their status. While these considerations would have a quantitative impact, the qualitative results would not be affected.

The relevant form of equation (1) for our problem is

$$\frac{1}{2}\sigma^2 V^2 F_{vv} + rVF_v - rF + F_t + \sum_{j=1}^n c_j \delta(t - t_j) = 0 \quad (12)$$

where c_j is the j th interest payment, t_j is the time at which the j th interest payment is made, n is the total number of interest payments, and $\delta(\cdot)$ is the Dirac delta function. The first derivative term does not involve the outflow of interest or dividend payments because they are exactly offset by the inflow of new financing. The standard terminal condition and the stopping condition described above complete the specification of the problem.

The solution can be obtained by the recursive technique discussed in the first section. For example, consider the situation immediately before the last payment is due. Let $s(V, t_n)$ be the value of the firm's stock if the payment is made. This is the solution to the standard problem with terminal condition $\max(V - P, 0)$. Then the minimum value of the firm at which the payment can be made, $\bar{V}$, is the root of $s(V, t_n) = c$. The value of the stock just before the payments is made, $\bar{s}(V, t_n)$, will be $s(V, t_n) - c$ if $V \geq \bar{V}$ and zero if $V < \bar{V}$. The value of the bonds will be $V - \bar{s}(V, t_n)$. For the situation just before the next-to-last payment is due, we apply the same analysis with $\bar{s}(V, t_n)$ replacing $\max(V - P, 0)$. By working recursively in this way, we can obtain a complete solution to the problem, but in general no closed form expression will be available.

To obtain a better perspective on the behavior of F , consider the case of a perpetual bond with continual interest payments of c per unit time. Equation (1) now has the form

$$\frac{1}{2}\sigma^2 V^2 F_{vv} + rVF_v - rF + c = 0. \quad (13)$$

From our earlier discussion we know that there will be some point at which no more equity can be sold and the bondholders will take over the firm. To find this point, think of things in the following way. In equilibrium new equity financing must sell at a fair price, so it makes no difference whether we think of it as being

purchased by new investors or by the original stockholders. So we can think of this as a situation where the stockholders will make payments into the firm to cover the interest payments to the bondholders, but at any time they have the right to stop making payments and either turn the firm over to the bondholders or pay them c/r . It is clear that the critical value of the firm at which they will do this, $\bar{V}$, is independent of the current value of the firm and will be chosen by the stockholders to minimize the value of the bonds and hence maximize the value of their own claim.

While a solution could be obtained and interpreted by the probabilistic approach discussed earlier, in the perpetual case it may be clearer to proceed formally with the ordinary differential equation (13). The solution to (13) can be written as the sum of a particular solution to the full inhomogeneous equation and the general solution to the corresponding homogeneous equation. A particular solution is c/r . Combining this with the corresponding general solution gives

$$F(V) = \frac{c}{r} + K_1 V + K_2 V^{-\alpha}, \quad (14)$$

where $\alpha = 2r/\sigma^2$ and K_1 and K_2 are arbitrary constants to be determined by the boundary conditions. As the value of the firm goes to infinity, the bonds must approach their riskless value and further increases in value must accrue solely to the stockholders, so $F_v(\infty) = 0$ and hence $K_1 = 0$.

The lower boundary condition then gives

$$K_2 \bar{V}^{-\alpha} + \frac{c}{r} = \min\left(\bar{V}, \frac{c}{r}\right), \quad (15)$$

so $K_2 = \bar{V}^{\alpha+1} - (c/r)\bar{V}^\alpha$ if $\bar{V} < c/r$ and $K_2 = 0$ if $\bar{V} \geq c/r$. Choosing $\bar{V} \geq c/r$ gives the bonds their maximum possible value, so the optimal $\bar{V}$ must be an interior point and the value of the bonds will be

$$F(V) = \frac{c}{r} + \left(\bar{V}^{\alpha+1} - \frac{c}{r}\bar{V}^\alpha\right)V^{-\alpha}. \quad (16)$$

Solving the first order condition for minimizing $F(V)$ gives $\bar{V} = (\alpha/\alpha+1)c/r$. Substitution and rearranging then gives

$$F(V) = \frac{c}{r} - \left[\left(\frac{\alpha}{\alpha+1}\right)^\alpha - \left(\frac{\alpha}{\alpha+1}\right)^{\alpha+1}\right]\left(\frac{c}{r}\right)^{\alpha+1} V^{-\alpha}. \quad (17)$$

For comparison consider now the corresponding case where the assets of the firm can be sold to make interest and dividend payments. The valuation equation for the bonds, G , will take the form

$$\frac{1}{2}\sigma^2 V^2 G_{vv} + [(r-a)V - (c+d)]G_v - rG + c = 0 \quad (18)$$

where c again represents the continuous interest payments to the bonds and the stock is entitled to receive dividend payments of $aV + d$. The upper boundary

condition will again be $F_v(\infty)=0$ and the lower boundary condition is now $F(0)=0$. The solution is

$$G(V) = \frac{c}{r} \left[1 - \frac{\Gamma\left(k - \frac{2(r-a)}{\sigma^2} + 2\right) \left(\frac{2(c+d)}{\sigma^2 V}\right)^k}{\Gamma\left(2k - \frac{2(r-a)}{\sigma^2} + 2\right)} \times M\left(k, 2k - \frac{2(r-a)}{\sigma^2} + 2, -\frac{2(c+d)}{\sigma^2 V}\right) \right], \quad (19)$$

where $M(\cdot, \cdot, \cdot)$ is the confluent hypergeometric function, $\Gamma(\cdot)$ is the gamma function, and k is the positive root of

$$\sigma^2 k^2 + [\sigma^2 - 2(r-a)]k - 2r = 0. \quad (20)$$

When $a=0$, $k=(2r/\sigma^2)=\alpha$, so with $d=0$ this reduces to formula (42) in Merton [9]. In this case (19) can be written in the more convenient form

$$G(V) = \frac{c}{r} [1 - \Gamma(\alpha, Z)] + \left(\frac{cV}{c+d}\right) \Gamma(\alpha+1, Z), \quad (21)$$

where $Z=(2(c+d)/\sigma^2 V)$ and $\Gamma(n, x)$ is the gamma distribution function with parameter n , $\Gamma(n, x) = \int_0^x e^{-s} s^{n-1} ds / \Gamma(n)$.⁹

8. Let $Z=(2(c+d)/\sigma^2 V)$ and $G(V)=Z^k e^{-Z} h(Z)$. This reduces the homogeneous part of the equation to

$$Zh_{zz} + [(\beta+k) - Z]h_z - \beta h = 0,$$

where $\beta = k - (2(r-a)/\sigma^2) + 2$. This is Kummer's equation, with general solution

$$K_1 M(\beta, \beta+k, Z) + K_2 Z^{1-\beta-k} M(1-k, 2-\beta-k, Z).$$

Using the boundary conditions and well-known properties of the confluent hypergeometric function gives (19).

9. The solution in this form was shown to us by John Barry. It has also been independently derived by Jonathan Ingersoll. That it is equivalent to the solution given by Merton can be seen by noting that

$$M(A, A+1, -Z) = AZ^{-A} \Gamma(A) \Gamma(A, Z)$$

and

$$\begin{aligned} M(A, A+2, -Z) &= (A+1)M(A, A+1, -Z) - AM(A+1, A+2, -Z) \\ &= \Gamma(A+2)Z^{-A} \Gamma(A, Z) \\ &\quad - A\Gamma(A+2)Z^{-(A+1)} \Gamma(A+1, Z). \end{aligned}$$

Analysis of the solutions shows that F is always greater than G , so the financing restrictions do increase the value of the bonds. When V is large, F is less sensitive to changes in V than is G , and it is less risky in the sense of having a lower elasticity, but when V is small the relationships are reversed. The premium due to the restrictions achieves its maximum at $\bar{V}$ and is a decreasing convex function of V . For the case with financing restrictions, we find that the value at which the stockholders would abandon the firm is a linear increasing function of c and a decreasing convex function of σ^2 and r .

We would suspect that the premium of F over G is due partly to the increase in asset value from the inflow of new financing and partly from the implicit safety covenant which places the firm in the hands of the bondholders at some positive value. To get some idea of the different effects, consider a bond, H , which allows the sale of assets but which has a safety covenant giving the bondholders control of the firm at $\bar{V}$. It is easy to verify that

$$H(V) = \frac{c}{r} + \lambda \left[G(V) - \frac{c}{r} \right], \quad (22)$$

where

$$\lambda = \left[\frac{\bar{V} - \frac{c}{r}}{G(\bar{V}) - \frac{c}{r}} \right].$$

Inspection shows that $F \geq H > G$. At $\bar{V}$, F and H have the same value by construction. As V increases the spread between them at first widens and then narrows to zero as the value of each claim approaches that of riskless debt, c/r . The sensitivity and riskiness of F compared to H is qualitatively the same as its comparison to G .

Further examination of the functions shows that both F and G are increasing concave functions of V and c . They are both decreasing functions of σ^2 , having an initial concave segment followed by a convex segment. Similarly, both elasticities are increasing functions of V , c , and σ^2 .

V. CONCLUSION

In this paper we first discussed some general issues in the valuation of contingent claims. We outlined some solution methods which could be applied even when the problem possesses inherent discreteness and discussed an intuitive way of interpreting the solutions. We then investigated the effects of three specific provisions often found in bond indentures. These were safety covenants, subordination arrangements, and restrictions on the financing of interest and dividend payments. We found that these provisions do indeed increase the value of bonds, and that they may have a quite significant effect on the behavior of the firm's securities.

The most important qualifications to our results involve the assumptions about the absence of bankruptcy costs and about the probabilistic process governing the value of the firm. Most of our general results should hold for other stochastic

processes, but of course the specific formulas and quantitative impact would be different. It should be noted that if the value of the firm follows a jump process, the value of a safety covenant may be drastically altered since the value of the firm could then reach points below the bankruptcy level without first passing through it.

The introduction of bankruptcy costs might have a more important effect. This would depend on the specific form of the bankruptcy costs and also on the influence of other factors, such as taxes, which would have to be introduced into the analysis to justify the existence of debt in a world with positive bankruptcy costs. However, their impact on our analysis should not be exaggerated. We are considering bankruptcy as simply the transfer of the entire ownership of the firm to the bondholders. The physical activities of the firm need not be affected. The bondholders may not want to actively run the company, but probably the stockholders did not either. The bondholders could retain the old managers or hire new ones, or they could refinance the firm and sell all or part of their holdings. Certain legal costs may be involved in the act of bankruptcy, but if contracts are carefully specified in the first place with an eye toward minimizing these costs, then their importance may be significantly reduced.

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